A variable is said to be qualitative if it is not measurable and cannot be expressed in specific units of measurement (numbers). This variable is also called categorical variable.

The variable height is a quantitative variable since it is measurable and is expressed in a number while the variable domicile is qualitative since it is not measured and is expressed as rural or urban. It is noted that they are free from units of measurement.

Quantitative Data:

Quantitative data (variable) are measurements that are collected or recorded as a number. Apart from the usual data like height, weight etc.,

Qualitative Data:

Qualitative data are measurements that cannot be measured on a natural numerical scale. For example, the blood types are categorized as O, A, B along with the Rh factors. They can only be classified into one of the pre assigned or pre designated categories.

1.7 Measurement Scales

There are four types of data or measurements scales called nominal, ordinal, interval and ratio. These measurement scale is made by Stanley Stevens.

1.7.1 Nominal scales:

Nominal measurement is used to label a variable without any ordered value. For example, we can ask in a questionnaire 'What is your gender? The answer is male or female. Here gender is a nominal variable and we associate a value 1 for male and 2 for a female.'

They are numerical for name sake only. For example, the numbers 1,2,3,4 may be used to denote a person being single, married, widowed or divorced respectively. These numbers do not share any of the properties of numbers we deal with in day to life. We cannot say $4 > 1$ or $2 < 3$ or $1+3 = 4$ etc. The order of listings in the categories is irrelevant here. Any statistical analysis carried out with the ordering or with arithmetic operations is meaningless.

1.7.2 Ordinal scales:

These data share some properties of numbers of arithmetic but not all properties. For example, we can classify the cars as small, medium and big depending on the size.

In the ordinal scales, the order of the values is important but the differences between each one is unknown. Look at the example below.

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